



Modelling Acute-Phase HIV Transmission in Complex Networks

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Problem Statement

HIV prevention: With 44.1 million lives lost and 39 million people currently living with the virus, HIV remains one of the world's most persistent public health challenges_[1]. Pre-Exposure Prophylaxis (PrEP) is >90% _{[2][3]} effective in preventing HIV contraction. However, universal therapy is economically unfeasible_[4].

Challenge: HIV transmission is driven by complex dynamics, including superspreaders and bisexual bridging, which are not included in standard epidemiological models. Policy-makers must identify which group of the population (e.g., male/female bisexuals, heterosexuals, gays) must be targeted to minimise HIV incidence. To address this, we developed a stochastic model that simulates complex dynamics, enabling a comparison of PrEP admission across different sub-populations.

Model Architecture



Network Initialization

- Input: Egodyads Dataset + Rahman et al. distributions
- Sampling of degrees (partners)
- Output: Static sexual contact network $G(V, E)$

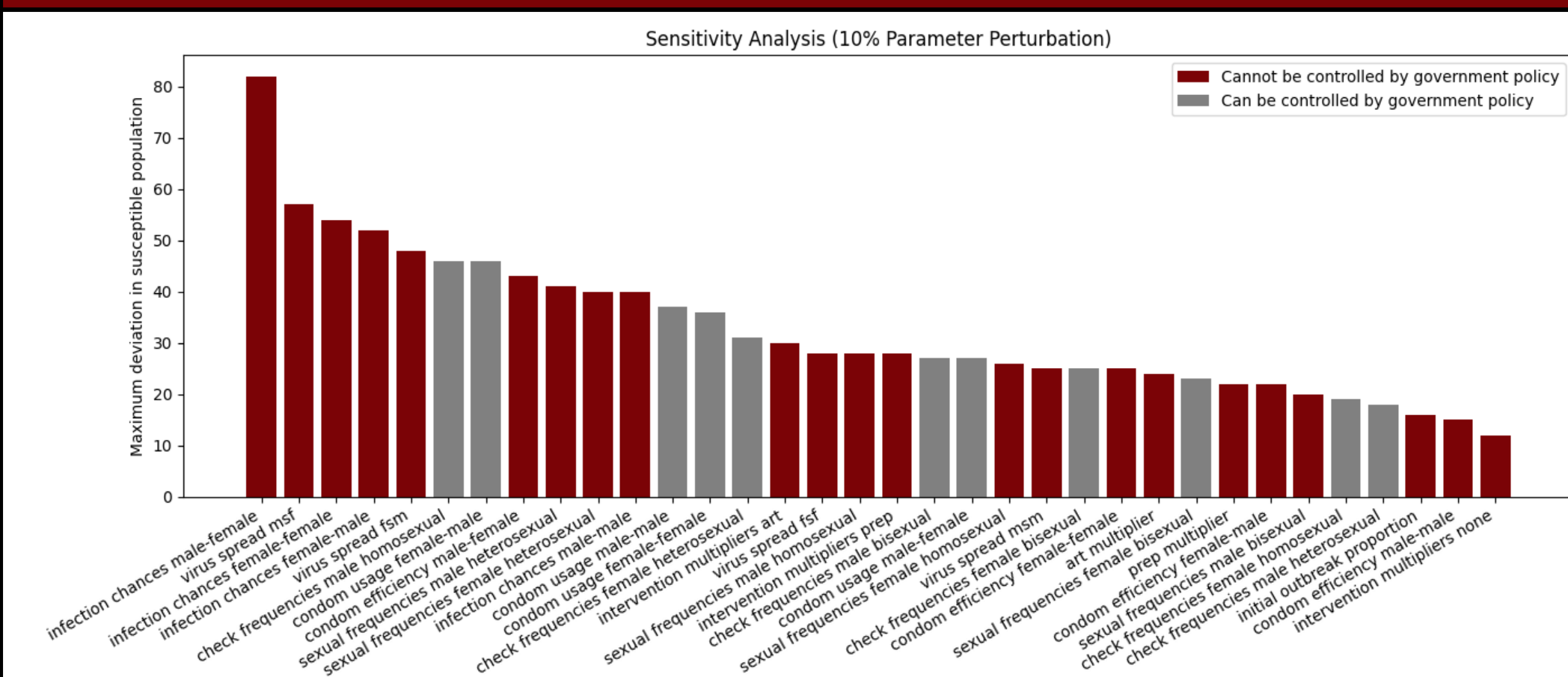
Stochastic Transmission Dynamics

- $P(\text{inf})_{i,j} = \omega_{i,j} \times \beta_{g_i, g_j} \times M_{\text{stage}} \times M_{\text{intervention}}$
- $\omega_{i,j}$: Frequency of sexual intercourse per week
- β_{g_i, g_j} : Infection probability per gender combination
- M_{stage} : Multipliers per infection stage
- $M_{\text{intervention}}$: Multipliers per intervention method

Temporal State Updates

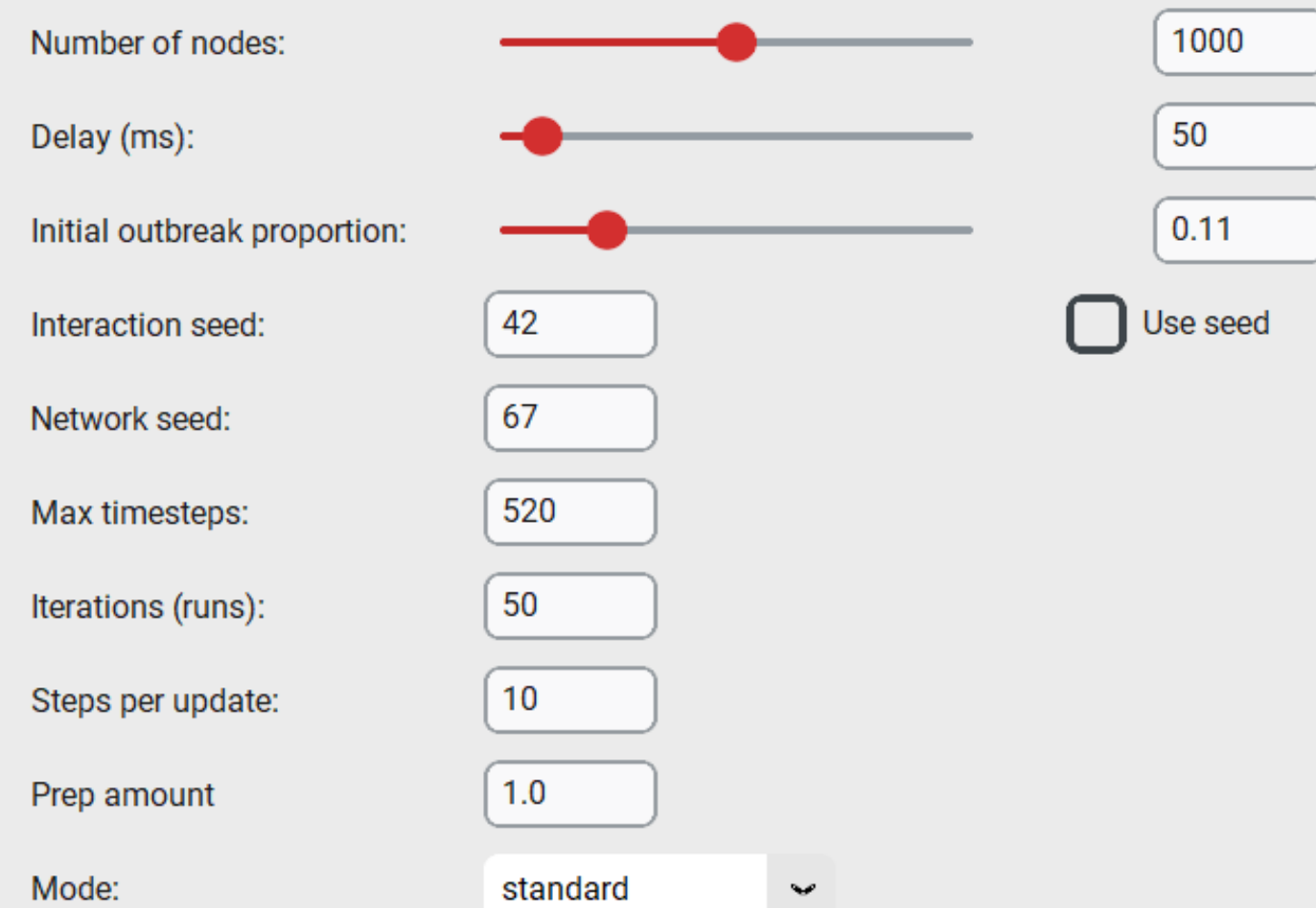
- Progression of time and disease phase: acute -> chronic -> AIDS -> death

Model Performance

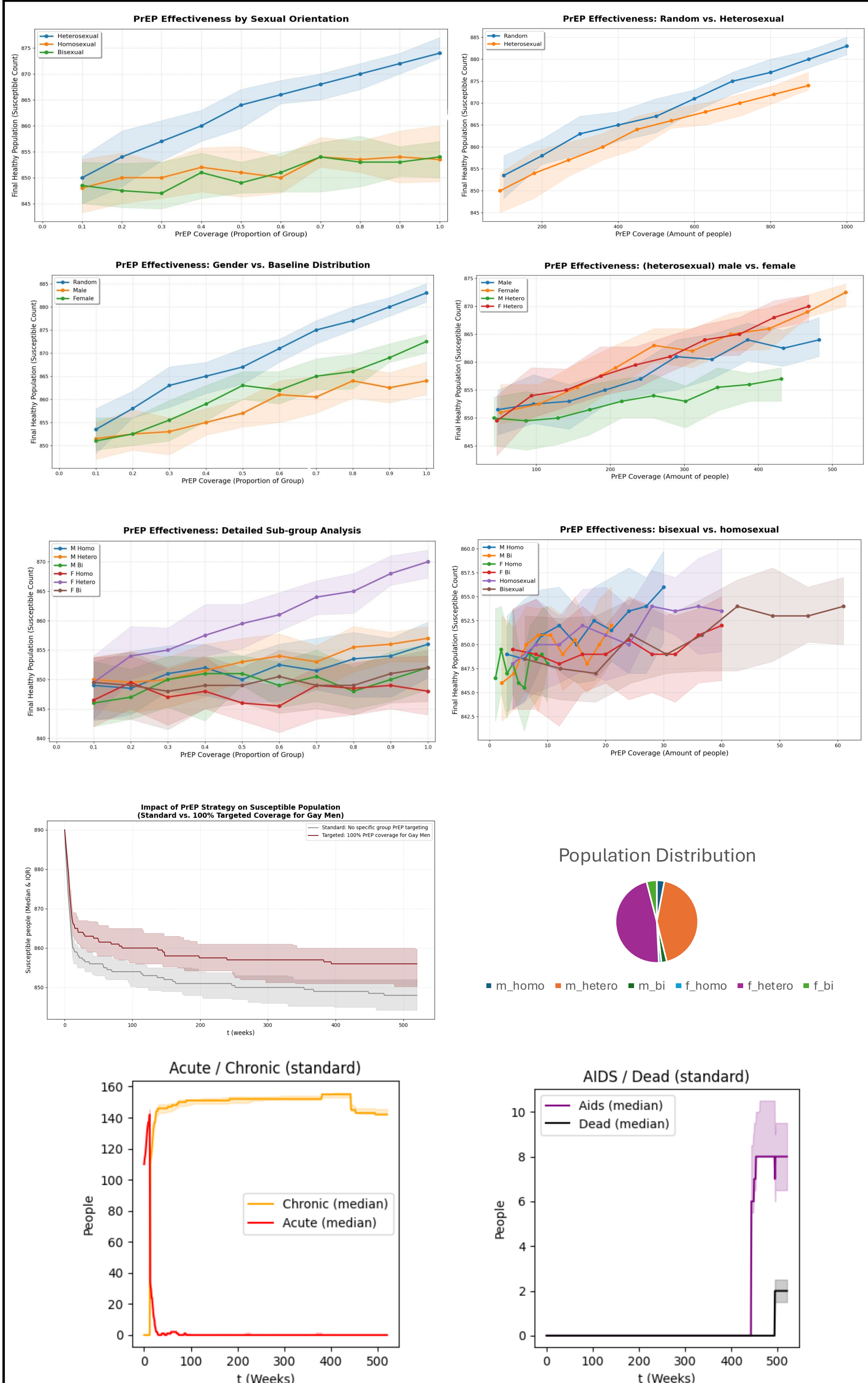


The sensitivity analysis shows that model outcomes are most influenced by infection probabilities, while policy-controllable interventions (testing and condom usage) have moderate effect.

Variables



Experimental Results



References

- [1] UNAIDS. (2023). Fact sheet — Global HIV & AIDS statistics. <https://www.unaids.org/en/resources/fact-sheet>
- [2] McCormack, S., Dunn, D. T., Desai, M., Dolling, D. I., Gafos, M., Gilson, R., ... & Gitt, O. N. (2016). Pre-exposure prophylaxis to prevent the acquisition of HIV-1 infection (PROUD): effectiveness results from the pilot phase of a pragmatic open-label randomised trial. The Lancet, 387(10013), 53-60.
- [3] CDC. (2022). Pre-Exposure Prophylaxis (PrEP). Centers for Disease Control and Prevention. <https://www.cdc.gov/hiv/risk/prap/index.html>
- [4] World Health Organization. (2022). Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: recommendations for a public health approach.

Check out the complete code and paper



Conclusion

Having male homosexuals take PrEP is the most effective and cost-efficient way to minimize HIV-spread via sexual network.